

Solving the 3D acoustic wave equation on generalized structured meshes: A finite-difference time-domain approach

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ABSTRACT

The key computational kernel of most advanced 3D seismic imaging and inversion algorithms used in exploration seismology involves calculating solutions of the 3D acoustic wave equation, most commonly with a finite-difference time-domain (FDTD) methodology. Although well suited for regularly sampled rectilinear computational domains, FDTD methods seemingly have limited applicability in scenarios involving irregular 3D domain boundary surfaces and mesh interiors best described by non-Cartesian geometry (e.g., surface topography). Using coordinate mapping relationships and differential geometry, an FDTD approach can be developed for generating solutions to the 3D acoustic wave equation that is applicable to generalized

3D coordinate systems and (quadrilateral-faced hexahedral) structured meshes. The developed numerical implementation is similar to the established Cartesian approaches, save for a necessary introduction of weighted first- and mixed second-order partial-derivative operators that account for spatially varying geometry. The approach was validated on three different types of computational meshes: (1) an “internal boundary” mesh conforming to a dipping water bottom layer, (2) analytic “semi-orthogonal cylindrical” coordinates, and (3) analytic semiorthogonal and numerically specified “topographic” coordinate meshes. Impulse response tests and numerical analysis demonstrated the viability of the approach for kernel computations for 3D seismic imaging and inversion experiments for non-Cartesian geometry scenarios.

INTRODUCTION

Calculating numerical solutions of the 3D acoustic wave equation represents the key computational kernel of most advanced 3D seismic imaging and inversion methods currently used in seismic exploration. These algorithms include the high-end 3D seismic techniques of reverse time migration (RTM), full-waveform inversion (FWI), and wave-equation migration velocity analysis (WEMVA). Finite-difference time-domain (FDTD) methods represent the most common numerical approach for generation solutions of the 3D acoustic wave equation, and have long been developed and widely implemented since the 1980s (Robertsson et al. [2012] and references therein). Although the 3D FDTD modeling of acoustic wavefields is fairly straightforward for regularly sampled rectilinear meshes, these approaches have seemingly lower applicability for scenarios in which seismic data are acquired on surfaces exhibiting irregular topology. Undertaking full-wavefield imaging and velocity inversion experiments in these situations requires handling irregular computa-

tional domains that are arguably best described by a more general non-Cartesian geometry. These types of scenarios may be found at many scale lengths, ranging from laboratory imaging experiments using ultrasonic measurements on cylindrical core plugs to seismic exploration scale over free-surface topography or irregular water-bottom surfaces.

One strategy for handling scenarios exhibiting irregular geometry is to turn to integral-based methods that solve the 3D acoustic wave equation on voxelized elements designed to conform to undulating domain boundary surfaces and to infill computational mesh interiors. Examples of these types of approaches include the finite-element (Marfurt, 1984), spectral-element (Komatitsch and Vilotte, 1998), and discontinuous-Galerkin (Cockburn et al., 2000) methods. By using meshes conformal to irregular surfaces, these approaches facilitate the numerical implementation of free-surface boundary conditions (BCs) (Priolo et al., 1994) and are able to more accurately represent discontinuities across major interior lithologic

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boundaries relative to finite-difference methods (Fornberg, 1988; Symes and Vdovina, 2009).

Although integral-based methods provide highly accurate — though computationally expensive — solutions to the 3D acoustic wave equation, herein, I argue that automatically precluding FDTD strategies for scenarios exhibiting irregular geometry and generalized (i.e., quadrilaterally faced hexahedral) structured meshes represents a severe and potentially unnecessary restriction. Rather, applying an FDTD methodology on irregular structured meshes is technically feasible and, in many cases, represents a cost-effective computational strategy. A successful implementation, though, requires addressing three key challenges: (1) how to generate a 3D computational domain that conforms to the desired geometry and exhibits favorable mesh characteristics, (2) how to specify the 3D acoustic wave equation appropriate for the irregular computational domain, and (3) how to provide a stable and numerically accurate solution of the 3D acoustic wave equation on that particular generalized 3D mesh.

Solving partial differential equations on irregularly shaped computational domains using coordinate transformation approaches and finite-difference operators is a common technique for boundary value problems in many science and engineering fields. Although these techniques have found a moderate amount of traction for modeling wavefield solutions of 2D/3D (visco-)elastic wave equations (Ohminato and Chouet, 1997; Hestholm, 1999; Hestholm and Ruud, 2002; Ely et al., 2008, 2009; Appellö and Petersson, 2009; Zhang et al., 2012), they are seldom used to solve the two-way acoustic wave equation for 3D full-wavefield imaging and velocity inversion problems, though a few examples exist in the literature for specific symmetric geometries (Nichols, 1994; Cheng and Blanch, 2008). The aim of the present work is to formulate and implement an $O(\Delta t^2, \Delta x^8)$ FDTD solution of the two-way acoustic wave equation that is analogous to $O(\Delta t^2, \Delta x^8)$ FDTD operators commonly used for performing wave propagation in Cartesian geometries, but is applicable to 3D computational domains exhibiting more generalized (i.e., nonsymmetric and nonorthogonal) geometry.

The acoustic wavefield propagation methodology advocated herein represents an adaptation of the Riemannian wavefield extrapolation (RWE) coordinate transform approach that uses differential geometry relationships and known mappings between 3D generalized and Cartesian coordinate systems to specify the Laplacian differential operator governing 3D acoustic wave propagation in that generalized coordinate system. Sava and Fomel (2005) present a derivation of RWE for acoustic wave propagation in 2D “semi-orthogonal” “Riemannian coordinates” and implement a one-way operator appropriate for 2D wave-equation migration (WEM). Shragge (2008) extends this approach to 3D nonorthogonal coordinates and similarly implements a one-way operator for 3D WEM imaging. Shragge (2014) demonstrates the applicability of this method for two-way acoustic wave equations and applies the developed FDTD operators to perform 2D RTM directly from topographic surfaces. Building on from these approaches, the goals of the present work are threefold: (1) to provide an extension of RWE theory to the two-wave acoustic wave equation for generalized (i.e., nonorthogonal) 3D geometries, (2) to demonstrate that one can develop generalized-coordinate $O(\Delta t^2, \Delta x^8)$ FDTD operators that are (nearly) equivalent to standard $O(\Delta t^2, \Delta x^8)$ FDTD operators applicable on Cartesian meshes (e.g., Dablain, 1986), and (3) to highlight some of the potential uses of the method for

3D acoustic wave propagation problems exhibiting non-Cartesian geometry.

The paper begins by presenting the theory of the 3D (constant-density) acoustic wave equation in generalized 3D coordinate systems based on coordinate mapping and differential geometry relationships. I then detail a numerical FDTD solution of the 3D acoustic wave equation that takes into account the spatial variability of non-Cartesian geometry. This leads to a discussion on the numerical considerations of stability and sufficiency of wavefield sampling. I tested the developed theory and the numerical scheme on three different computational meshes: (1) an “internal boundary” (IB) mesh conforming to a dipping water bottom, (2) analytic “semiorthogonal cylindrical” coordinates, and (3) analytic semiorthogonal and numerically specified “topographic” coordinate meshes. The paper concludes with a brief discussion on the GPU-based implementation and the memory and computational complexity overheads of adopting the 3D FDTD acoustic wavefield modeling approach advocated herein.

3D GENERALIZED ACOUSTIC WAVE EQUATION

The (constant-density) acoustic wave equation in a 3D generalized coordinate system defined by variables $\xi = [\xi_1, \xi_2, \xi_3]$ is given by

$$\left[\nabla_{\xi}^2 - \frac{1}{v_{\xi}^2} \frac{\partial^2}{\partial t^2} \right] \mathcal{U}_{\xi} = \mathcal{F}_{\xi}, \quad (1)$$

where ∇_{ξ}^2 is a generalized Laplacian operator, v_{ξ} is a velocity field, $\mathcal{U}_{\xi}$ is a scalar acoustic wavefield, and $\mathcal{F}_{\xi}$ is the source distribution. Here and throughout, I use a notation where subscript ξ designate fields and operators defined in the generalized ξ -coordinate system, whereas those in Cartesian coordinate system $\mathbf{x} = [x_1, x_2, x_3]$ are specified by a subscript $\mathbf{x}$. Cartesian $\mathbf{x}$ -coordinates represent an irregularly shaped *physical domain* over which one desires to calculate a solution to the 3D acoustic wave equation. Generalized ξ -coordinates represent a transformed regular *canonical domain* on which one actually computes an acoustic wavefield solution. The relationship between the ξ - and $\mathbf{x}$ -coordinate systems is assumed to be known, unique (i.e., one-to-one), and expressible through a set of analytic or numeric mapping equations $x_i = f_i(\xi_j)$, where $i, j = 1, 2, 3$ (Shragge, 2008, 2014).

Because the partial differential operators of the generalized Laplacian ∇_{ξ}^2 are specified in the ξ -coordinate system, they naturally are affected by spatially varying geometry. Providing the correct formulation of the Laplacian operator on the ξ -mesh involves introducing transformations from the mathematical field of differential geometry in the form of the (symmetric rank-two) metric tensor $[g_{ij}]$

$$[g_{ij}] = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{12} & g_{22} & g_{23} \\ g_{13} & g_{23} & g_{33} \end{bmatrix}, \quad (2)$$

whose elements g_{ij} provide a link between the ξ - and $\mathbf{x}$ -coordinate systems through

$$g_{ij} = \frac{\partial x_k}{\partial \xi_i} \frac{\partial x_k}{\partial \xi_j}, \quad \text{where } i, j = 1, 2, 3. \quad (3)$$

Unless stated otherwise, I use summation notation over repeated indices and a convention in which the subscript and superscript indices on matrices (e.g., g_{ij} and g^{ij}) indicate covariant and contravariant tensors, respectively (Synge and Schild, 1978).

Specifying the 3D acoustic wave equation in generalized coordinates requires using a contravariant representation of the metric tensor (i.e., g^{ij}) defined as the matrix inverse of the covariant metric tensor g_{ij}

$$[g^{ij}] = \frac{1}{|\mathbf{g}|} \begin{bmatrix} g_{22}g_{33} - g_{23}^2 & g_{13}g_{23} - g_{12}g_{33} & g_{12}g_{33} - g_{13}g_{22} \\ g_{13}g_{23} - g_{12}g_{33} & g_{11}g_{33} - g_{13}^2 & g_{12}g_{13} - g_{11}g_{23} \\ g_{12}g_{33} - g_{13}g_{22} & g_{12}g_{13} - g_{11}g_{23} & g_{11}g_{22} - g_{12}^2 \end{bmatrix}, \quad (4)$$

where the metric tensor discriminant $|\mathbf{g}|$ is given by

$$|\mathbf{g}| = |g_{11}g_{22}g_{33} - g_{12}^2g_{33} - g_{23}^2g_{11} - g_{13}^2g_{22} + 2g_{12}g_{13}g_{23}|. \quad (5)$$

Computing the partial derivatives of Laplacian operator ∇_{ξ}^2 in equation 1 involves introducing the contravariant metric tensor elements from equations 4 and 5 into a generalized expression for the Laplacian operator (Guggenheimer, 1977):

$$\nabla_{\xi}^2 = \frac{1}{\sqrt{|\mathbf{g}|}} \frac{\partial}{\partial \xi_i} \left(g^{ij} \sqrt{|\mathbf{g}|} \frac{\partial}{\partial \xi_j} \right), \quad i, j = 1, 2, 3, \quad (6)$$

where partial derivatives $\partial/\partial \xi_i$ are evaluated with respect to the generalized ξ -coordinate variables.

To facilitate the numerical solution of the generalized 3D acoustic wave equation below, I expand the generalized Laplacian operator into first- and second-order partial differential terms:

$$\begin{aligned} \nabla_{\xi}^2 &= \frac{1}{\sqrt{|\mathbf{g}|}} \frac{\partial}{\partial \xi_i} \left(g^{ij} \sqrt{|\mathbf{g}|} \right) \frac{\partial}{\partial \xi_j} + g^{ij} \frac{\partial^2}{\partial \xi_i \partial \xi_j} \\ &= \zeta^i \frac{\partial}{\partial \xi_i} + g^{ij} \frac{\partial^2}{\partial \xi_i \partial \xi_j}, \end{aligned} \quad (7)$$

where, for notational convenience, I define the purely geometric scalar fields ζ^i as

$$\begin{aligned} \zeta^i &\equiv \frac{1}{\sqrt{|\mathbf{g}|}} \frac{\partial \left(\sqrt{|\mathbf{g}|} g^{ij} \right)}{\partial \xi_j} \\ &= \frac{1}{\sqrt{|\mathbf{g}|}} \left[\frac{\partial \left(\sqrt{|\mathbf{g}|} g^{i1} \right)}{\partial \xi_1} + \frac{\partial \left(\sqrt{|\mathbf{g}|} g^{i2} \right)}{\partial \xi_2} + \frac{\partial \left(\sqrt{|\mathbf{g}|} g^{i3} \right)}{\partial \xi_3} \right]. \end{aligned} \quad (8)$$

The ζ^i fields are independent of the acoustic wavefield $\mathcal{U}_{\xi}$ and are measures of the rates by which space expands/compresses and/or shears in the i th direction, and can be nonzero even for common (e.g., polar, spherical) orthogonal coordinate systems (Shragge, 2008). Note that setting tensor elements $g^{ij} = \delta^{ij}$, where δ^{ij} is a

Kronecker delta function, yields the familiar Laplacian operator in Cartesian coordinates. Finally, substituting the expressions in equations 7 and 8 into equation 1 yields

$$\zeta^i \frac{\partial \mathcal{U}_{\xi}}{\partial \xi_i} + g^{ij} \frac{\partial^2 \mathcal{U}_{\xi}}{\partial \xi_i \partial \xi_j} = \frac{1}{v_{\xi}^2} \frac{\partial^2 \mathcal{U}_{\xi}}{\partial t^2} + \mathcal{F}_{\xi}, \quad (9)$$

which is the form of the 3D acoustic wave equation desired for the FDTD numerical implementation discussed below.

3D GENERALIZED FDTD PROPAGATION

The goal of this section is to develop a 3D FDTD propagation scheme appropriate for generating numerical solutions of the 3D acoustic wave equation on generalized structured meshes. Herein, I demonstrate that implementing a 3D FDTD approach in generalized coordinates is a straightforward undertaking that requires only two minor modifications to established 3D Cartesian FDTD approaches. First, one must account for the spatially varying metric tensor components when applying finite-difference (FD) stencils to discretized wavefields. Second, one needs to incorporate first- and mixed second-order partial derivatives, as well as the standard second-order derivatives, of the generalized Laplacian operator in equation 7.

Numerical scheme

The proposed FDTD scheme is based on FD approximations of order $O(\Delta \xi^8)$ for the spatial first- and second-order derivative operators and an $O(\Delta \xi^4)$ approximation of the mixed second-order partial derivatives (where $\Delta \xi$ is the sampling interval in the canonical domain). Continuous wavefields are approximated by a discrete representation, $\mathcal{U}_{\xi}(\xi, t) \approx \mathcal{U}_{l,m,n}^p$, where $l = [1, L]$, $m = [1, M]$ and $n = [1, N]$ are spatial indices in the ξ_1 , ξ_2 , and ξ_3 directions, respectively, and $p = [1, P]$ is the temporal index. Similarly, the continuous and time-independent ζ^i fields and g^{ij} metric tensor components are represented by $\zeta_{l,m,n}^i$ and $g_{l,m,n}^{ij}$. Given these representations and assuming that discretization of the computational mesh is uniform in all directions (i.e., $\Delta \xi \equiv \Delta \xi_1 = \Delta \xi_2 = \Delta \xi_3$), I rewrite the spatial first-order partial-derivative FD operators as

$$\zeta^1 \frac{\partial \mathcal{U}_{\xi}}{\partial \xi_1} \approx \frac{\zeta_{l,m,n}^1}{\Delta \xi} \sum_{i=1}^4 F_i [\mathcal{U}_{l+i,m,n}^p - \mathcal{U}_{l-i,m,n}^p], \quad (10)$$

$$\zeta^2 \frac{\partial \mathcal{U}_{\xi}}{\partial \xi_2} \approx \frac{\zeta_{l,m,n}^2}{\Delta \xi} \sum_{i=1}^4 F_i [\mathcal{U}_{l,m+i,n}^p - \mathcal{U}_{l,m-i,n}^p], \quad (11)$$

$$\zeta^3 \frac{\partial \mathcal{U}_{\xi}}{\partial \xi_3} \approx \frac{\zeta_{l,m,n}^3}{\Delta \xi} \sum_{i=1}^4 F_i [\mathcal{U}_{l,m,n+i}^p - \mathcal{U}_{l,m,n-i}^p], \quad (12)$$

the second-order partial-derivative FD operators as

$$g^{11} \frac{\partial^2 \mathcal{U}_\xi}{\partial \xi_1^2} \approx \frac{g_{l,m,n}^{11}}{\Delta \xi^2} \sum_{i=0}^4 S_i [\mathcal{U}_{l+i,m,n}^p + \mathcal{U}_{l-i,m,n}^p], \quad (13)$$

$$g^{22} \frac{\partial^2 \mathcal{U}_\xi}{\partial \xi_2^2} \approx \frac{g_{l,m,n}^{22}}{\Delta \xi^2} \sum_{i=0}^4 S_i [\mathcal{U}_{l,m+i,n}^p + \mathcal{U}_{l,m-i,n}^p], \quad (14)$$

$$g^{33} \frac{\partial^2 \mathcal{U}_\xi}{\partial \xi_3^2} \approx \frac{g_{l,m,n}^{33}}{\Delta \xi^2} \sum_{i=0}^4 S_i [\mathcal{U}_{l,m,n+i}^p + \mathcal{U}_{l,m,n-i}^p], \quad (15)$$

and the mixed second-order partial-derivative FD operators as

$$g^{12} \frac{\partial^2 \mathcal{U}_\xi}{\partial \xi_1 \partial \xi_2} \approx \frac{g_{l,m,n}^{12}}{\Delta \xi^2} \sum_{j=1}^2 \sum_{i=1}^2 M_{ij} [\mathcal{U}_{l+i,m+j,n}^p + \mathcal{U}_{l-i,m-j,n}^p - \mathcal{U}_{l+i,m-j,n}^p - \mathcal{U}_{l-i,m+j,n}^p], \quad (16)$$

$$g^{13} \frac{\partial^2 \mathcal{U}_\xi}{\partial \xi_1 \partial \xi_3} \approx \frac{g_{l,m,n}^{13}}{\Delta \xi^2} \sum_{j=1}^2 \sum_{i=1}^2 M_{ij} [\mathcal{U}_{l+i,m,n+j}^p + \mathcal{U}_{l-i,m,n-j}^p - \mathcal{U}_{l+i,m,n-j}^p - \mathcal{U}_{l-i,m,n+j}^p], \quad (17)$$

$$g^{23} \frac{\partial^2 \mathcal{U}_\xi}{\partial \xi_2 \partial \xi_3} \approx \frac{g_{l,m,n}^{23}}{\Delta \xi^2} \sum_{j=1}^2 \sum_{i=1}^2 M_{ij} \times [\mathcal{U}_{l,m+i,n+j}^p + \mathcal{U}_{l,m-i,n-j}^p - \mathcal{U}_{l,m+i,n-j}^p - \mathcal{U}_{l,m-i,n+j}^p]. \quad (18)$$

The F_i , S_i , and M_{ij} used in equations 10–18 are FD coefficients, the values of which used herein are found in Table 1. For the second-order temporal partial derivative term, I use a standard $O(\Delta t^2)$ approximation,

$$\frac{1}{v_\xi^2} \frac{\partial^2 \mathcal{U}_\xi}{\partial t^2} \approx \frac{1}{v_{l,m,n}^2 \Delta t^2} [\mathcal{U}_{l,m,n}^{p+1} - 2\mathcal{U}_{l,m,n}^p + \mathcal{U}_{l,m,n}^{p-1}]. \quad (19)$$

Table 1. Stencil FD coefficients F_i , S_i , and M_{ij} used in equations 10–18.

Coefficient	Value	Coefficient	Value	Coefficient	Value
F_0	—	S_0	$-\frac{205}{72}$	M_{11}	$\frac{4}{9}$
F_1	$\frac{4}{5}$	S_1	$\frac{8}{5}$	M_{12}	$-\frac{1}{18}$
F_2	$-\frac{1}{5}$	S_2	$-\frac{1}{5}$	M_{21}	$-\frac{1}{18}$
F_3	$\frac{4}{105}$	S_3	$\frac{8}{315}$	M_{22}	$\frac{1}{144}$
F_4	$-\frac{1}{280}$	S_4	$-\frac{1}{560}$	—	—

Inserting the expressions in equations 10–19 into equation 9 and solving for the $\mathcal{U}_{i,j,k}^{p+1}$ term leads to a 3D FDTD propagation scheme for modeling the temporal evolution of an acoustic wavefield on a generalized coordinate mesh.

Figure 1a shows the 3D FD stencil applied at each $[l, m, n]$ mesh point to compute wavefield value $\mathcal{U}_{l,m,n}^{p+1}$. The mixed second-order FD operators, which arise because of the nonorthogonal geometry, introduce an additional 48 points into the stencil relative to the conventional $O(\Delta \xi^8)$ 25-point FD “star” shown in Figure 1b. Although this represents a 2.92× increase in the number of stencil points, some of the extra computational complexity may be hidden through appropriate use of L1 cache memory, coalesced memory calls and GPU-based implementation optimizations (see the section “Discussion”).

Finally, because enabling the computation of acoustic wavefields on meshes conforming to irregular surfaces is a main motivation of the present work, it is appropriate to discuss one of the key advantages using the above generalized coordinate FDTD scheme: implementing the free-surface BC. Recall that a key challenge of Cartesian-based modeling of acoustic wavefields on domains exhibiting irregular topography is accurately applying the free-surface BC. However, by using a topography-conforming coordinate transformation to a regular computational mesh one can effectively recast the “topographic free-surface BC” problem back to a “Cartesian-like free-surface BC” problem — provided the correct geometric factors are specified. To generate the examples, herein, I use a reflecting free-surface BC; however, I assert that other BCs — e.g., the absorbing BC (ABC) or the perfectly matched layer (PML) — can be specified for the transformed computational mesh and implemented in a fairly straightforward fashion.

Numerical considerations: Stability

The stability of the approach for the 3D FDTD scheme described above is controlled by the discretizations $\Delta \xi_i$ in the canonical ξ -coordinate domain. However, because metric coefficients g^{ij} are in general nonstationary throughout the ξ -mesh, the coordinate system distortion as viewed in the physical domain must be taken into account when considering discretization and associated numerical stability. Herein, I do provide a rigorous derivation of a Courant-Frederich-Lewy (CFL) criterion of the 3D FDTD approach for generalized coordinate geometry (a von Neumann analysis in the Fourier domain is not possible because of the nonstationarity of the coefficients). Rather, I derive a heuristic sufficiency condition based on the CFL criterion derived for FDTD propagation schemes for the Cartesian 3D acoustic wave equation in a homogeneous medium. For these scenarios, the upper bound on the maximum allowable time step Δt is given by (Lines et al., 1999)

$$\Delta t \leq \frac{\Delta r_H}{v_x}, \quad (20)$$

where v_x is the velocity in Cartesian physical coordinates and Δr_H is the harmonic root-mean-square (rms) of the spatial sampling intervals given by

$$\Delta r_H = [\Delta x_1^{-2} + \Delta x_2^{-2} + \Delta x_3^{-2}]^{-\frac{1}{2}}, \quad (21)$$

where Δx_i are the discretization intervals along the three coordinate axes.

For generalized geometries the effective discretization intervals necessarily become functions of the canonical ξ -coordinate system itself (i.e., $\Delta x_i = \Delta x_i(\xi_j)$), which leads to a spatially varying CFL criterion. Assuming that as $\Delta x_i \rightarrow 0$: (1) one may represent a discretized element by a differential $\Delta x_i \rightarrow dx_i$ and (2) a total derivative approximation (i.e., $dx_i = \partial x_i / \partial \xi_j d\xi_j$) may be substituted for Δx_i :

$$\Delta x_i \approx \frac{\partial x_i}{\partial \xi_j} \Delta \xi_j, \quad (22)$$

one may rewrite the CFL criterion in equation 20 as

$$\Delta t \leq \frac{1}{v_{\xi}} \arg \min_{\xi_1, \xi_2, \xi_3} \left[\left(\frac{\partial x_1}{\partial \xi_j} \Delta \xi_j \right)^{-2} + \left(\frac{\partial x_2}{\partial \xi_j} \Delta \xi_j \right)^{-2} + \left(\frac{\partial x_3}{\partial \xi_j} \Delta \xi_j \right)^{-2} \right]^{-\frac{1}{2}}, \quad (23)$$

where $\arg \min$ is used to enforce the condition that time step Δt must be lower than the global minimum of the reciprocal-velocity-weighted Δr_H function over the entire ξ -domain. Note that the expression for Δr_H for a uniformly sampled Cartesian mesh (i.e., $\Delta \xi = \Delta x$) simplifies to $\Delta r_H = \Delta x / \sqrt{3}$ and yields

$$\Delta t \leq \frac{\Delta x}{\sqrt{3} v_{\xi}}, \quad (24)$$

which is the standard CFL criterion for 3D FDTD numerical schemes for the 3D acoustic wave equation.

Because the maximum allowable time step introduced by the CFL criterion is dependent on the harmonic rms of the spatial sampling intervals, the volume of the voxels in which the coordinate system mesh is most compressed will be the main constraint on

the maximum allowable time stepping interval and thus the overall computational complexity and runtime of the 3D FDTD propagation scheme. This suggests that one of the goals of a 3D computational mesh generation algorithm should be to prevent voxels from becoming too small in volume. (Alternatively, one could use adaptive time-stepping approaches to circumvent these issues; however, this extension is not discussed herein.)

Numerical considerations: Sampling

Simulating acoustic wave propagation on non-Cartesian meshes will also introduce spatial variation in points-per-wavelength (ppw) sampling. One may obtain a ppw estimate $P(\xi)$ via

$$P(\xi) \approx \frac{v_{\xi}}{\Delta r_{Ef}}, \quad (25)$$

where f is the frequency under consideration and Δr_E is the (Euclidean) rms distance estimate given by

$$\Delta r_E = \sqrt{\left(\frac{\partial x_1}{\partial \xi_j} \Delta \xi_j \right)^2 + \left(\frac{\partial x_2}{\partial \xi_j} \Delta \xi_j \right)^2 + \left(\frac{\partial x_3}{\partial \xi_j} \Delta \xi_j \right)^2}. \quad (26)$$

For Cartesian meshes with uniform sampling (i.e., $\Delta x = \Delta \xi$), this reduces to

$$P(\mathbf{x}) \approx \frac{v_{\mathbf{x}}}{\sqrt{3} \Delta x f}. \quad (27)$$

Note that because this function is controlled by the Euclidean rms norm, voxels significantly larger than the mean will have a lower

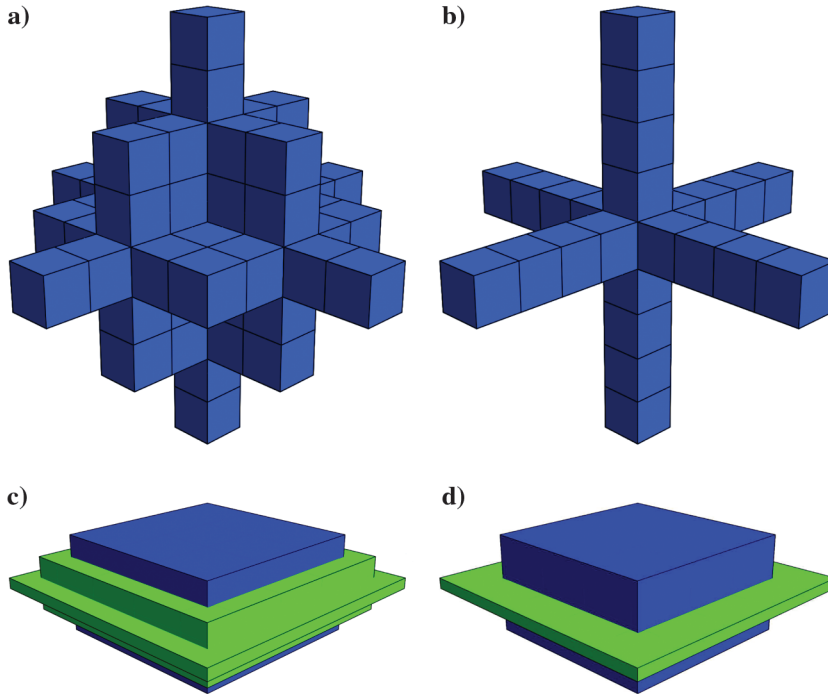


Figure 1. Visualization of FD stencils and grid blocks called in 3D FDTD method. (a) The 73-point FD stencil developed for 3D FDTD-based acoustic wave-propagation on nonorthogonal grids. (b) A common 25-point star FD stencil applied to Cartesian grids. (c) Grid of size $16 \times 16 \times 9$ (blue) with stencil halo points (green) loaded in to GPU cache memory during CUDA-based implementation for stencil in (a). (d) Grid of size $16 \times 16 \times 9$ (blue) with boundary halo region (green) loaded into the GPU cache memory during CUDA-based implementation of the 25-point stencil shown in (b).

ppw sampling and would be subject to greater dispersion error than smaller voxels. This suggests that a second computational meshing goal is to prevent voxels from becoming too large in volume.

NUMERICAL EXAMPLES

This section investigates 3D acoustic wave propagation on three different coordinate meshes: (1) IB coordinates conforming to an irregular seafloor surface, (2) analytic semiorthogonal cylindrical coordinates, and (3) analytic and numerically defined topographic coordinates. These computational meshes provide informative tests of the generalized 3D acoustic wave-equation theory and of the implementation of the 3D FDTD numerical scheme described above.

Internal boundary coordinate systems

One potential use of generalized 3D coordinate systems is to create meshes that are conformal to irregular internal boundaries, such as those at major lithologic interfaces (e.g., seafloor, geologic unconformities, etc.). By aligning the coordinate mesh with these boundaries one can reduce — or potentially eliminate — many of the deleterious effects of discretization (e.g., stair-casing commonly observed in Cartesian simulated wavefields from dipping interfaces). Similarly, this approach could also be used to improve algorithmic efficiency, such as allowing for wavefield injection of data from ocean bottom cable (OBC) or node (OBN) data sets along a single interface conformal to irregular sea-floor topography.

Generating meshes conformal to one or more generic parametric surfaces (i.e., $P_i(\xi_1, \xi_2, \xi_3 = \text{const.})$) can be accomplished by using Bézier interpolating functions (Farin, 1997). Herein, I present an example of a 3D coordinate system conformal to: (1) the free-surface boundary (i.e., $P_0(\xi_1, \xi_2, \xi_3 = 0) = 0$), (2) a uniformly flat layer at depth (i.e., $P_2(\xi_1, \xi_2, \xi_3 = 1) = 1$), and (3) an irregular water-bottom surface given by $P_1(\xi_1, \xi_2, \xi_3 = 0.5) = P(\xi_1, \xi_2)$. Importantly, in the context of this work, because the 3D IB coordinate system exactly conforms to a 3D Cartesian mesh at the $\xi_3 = 0$ and $\xi_3 = 1$ depth levels, this computational mesh allows for a direct evaluation of the numerical accuracy of the above 3D FDTD scheme relative to the equivalent Cartesian scheme, and thus represents an important numerical verification of the described approach.

A 3D IB coordinate system may be defined by three mapping equations:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3^2 + \xi_3(1 - \xi_3)(4P - 1) \end{bmatrix}, \quad (28)$$

where $P = P(\xi_1, \xi_2, \xi_3 = 0.5)$ is the field representing the depth to the water-bottom surface.

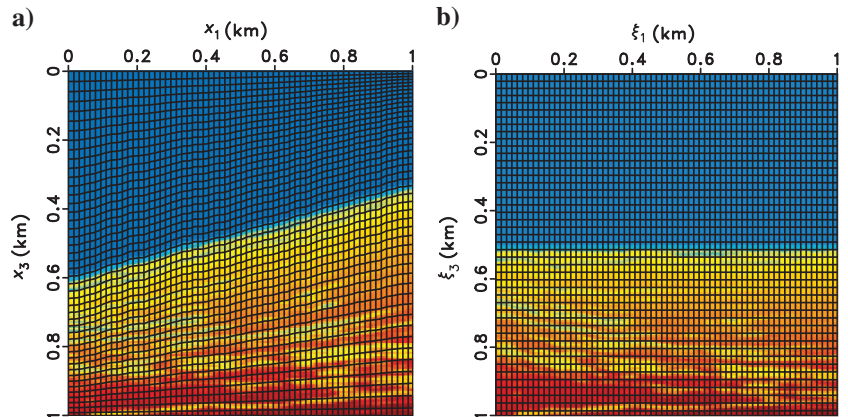
Figure 2 presents an example of the IB coordinate system defined in equation 28. Figure 2a shows a 2D cross section from a 3D velocity model that has a water-bottom layer dipping at a 15° angle. The velocity field ranges from 1.5 km s^{-1} in the water column (dark blue) to roughly 2.8 km s^{-1} in the sediment section (dark red). I have also overlain the IB-coordinate mesh that is conformal to the dipping water bottom (because the model is invariant in the third dimension, I have excluded it from this figure). Figure 2b shows the velocity model under the coordinate transformation mapping (i.e., pointwise interpolation) into the ξ -computational domain, which illustrates that the water bottom conforms to a single coordinate contour at $\xi_3 = 0.5 \text{ km}$.

As per equation 9, implementing generalized 3D acoustic wave propagation requires (pre)computing the g^{ij} and ζ^i field components. Although one may calculate these coefficients based on the numerical partial derivatives of the mapping relationships in equation 28, it is advantageous to compute these factors analytically wherever possible to forestall the introduction of numerical error. The analytic expression for the inverse metric tensor corresponding to the IB coordinate system in equation 28 can be derived after somewhat involved mathematical operations:

$$[g^{ij}] = \gamma^2 \begin{bmatrix} 1 & 0 & -4\gamma\xi_3(\xi_3 - 1)\frac{\partial P}{\partial \xi_1} \\ 0 & 1 & -4\gamma\xi_3(\xi_3 - 1)\frac{\partial P}{\partial \xi_2} \\ -4\gamma\xi_3(\xi_3 - 1)\frac{\partial P}{\partial \xi_1} & -4\gamma\xi_3(\xi_3 - 1)\frac{\partial P}{\partial \xi_2} & 1 + 16\xi_3^2(\xi_3 - 1)^2 \left[\left(\frac{\partial P}{\partial \xi_1} \right)^2 + \left(\frac{\partial P}{\partial \xi_2} \right)^2 \right] \end{bmatrix}, \quad (29)$$

where $\gamma = 1 - 4\xi_3 + P(8\xi_3 - 4)$, and the expression now includes partial derivatives of P in the ξ_1 and ξ_2 directions. The only nonzero ζ^i function is

Figure 2. A 3D IB-computational mesh specified using Bézier interpolating functions. Velocities range from 1.5 km s^{-1} (dark blue) to 2.8 km s^{-1} (dark red). (a) Velocity model in Cartesian coordinates x_1 - x_3 with dipping seafloor and an overlain IB coordinate mesh. (b) Velocity model from (a) interpolated to the IB ξ -coordinate system in the ξ_1 - ξ_3 -plane.



$$\zeta^3 = \frac{4\xi_3(\xi_3 - 1)}{\gamma} \left[\frac{\partial^2 P}{\partial \xi_1^2} + \frac{\partial^2 P}{\partial \xi_2^2} \right] + \frac{(8P - 4)(1 + 16\xi_3^2(\xi_3 - 1)^2)}{\gamma^3} \left[\left(\frac{\partial P}{\partial \xi_1} \right)^2 + \left(\frac{\partial P}{\partial \xi_2} \right)^2 \right] - \frac{32\xi_3(1 - 3\xi_3 + 2\xi_3^2)}{\gamma^2}. \quad (30)$$

Note that the expression takes into consideration the local curvature of the interior surface P through the second-order partial derivative operators, and that the coordinate system is semiorthogonal (i.e., ξ_1 and ξ_2 are mutually orthogonal but not to ξ_3) because $g^{12} = 0$. Inserting these expressions in equations 29 and 30 into equation 9 leads to the 3D acoustic wave equation and the corresponding 3D FDTD approximations for performing wavefield propagation on a 3D IB coordinate mesh.

I perform an impulse response (IR) test of the numerical 3D FDTD scheme using an IB mesh of dimension $N_\xi^3 = 320^3$ assuming a uniform sampling interval of $\Delta\xi = 3.125$ m, a constant $v_\xi = 2$ km s⁻¹ velocity, and a 25-Hz source wavelet injected at the central grid point on the free surface ($\xi = [0.5, 0.5, 0]$ km). Figure 3a shows the IR computed in 3D IB coordinates interpolated back to a Cartesian mesh, in which the black lines indicate the cut locations on each of the three faces. Although this result has been computed in IB coordinates, the interpolated IR is hemispherical and symmetric about the shot point as demanded by theory. Note also the absence of diffractions from the non-Cartesian grid. For reference, Figure 3b shows the IR computed directly in a Cartesian coordinate system.

To better examine the accuracy of the generalized coordinate 3D FDTD numerical scheme, I extract traces at three locations from the wavefields simulated through the IB and Cartesian coordinate systems. Figure 4 presents the trace-by-trace comparisons, in which the red and green lines show the Cartesian and IB coordinate simulation results, respectively, whereas the black line shows the waveform difference. Figure 4a presents the waveforms extracted at location $\mathbf{x} = [0.25, 0.25, 1]$ km. The first pulse is the direct arrival at 1-km depth, whereas the second and third pulses are from boundary reflections (i.e., reflecting BCs were used). Overall, the waveform fit is very good, though the Cartesian waveform exhibits a very slight delay with respect to the IB waveforms (i.e., by less than a single time sample) and thus does not provide a perfect match. Figure 4b and 4c presents similar results for locations $\mathbf{x} = [0.5, 0.5, 1.0]$ km and $\mathbf{x} = [0.75, 0.75, 1]$ km, respectively. Again, these panels illustrate very good matches between the waveforms computed in the two different coordinate systems.

I next demonstrate that the 3D FDTD approach is applicable for heterogeneous media by repeating the wavefield propagation tests using the velocity model shown in Figure 2. Figure 5a presents the wavefield computed through a 3D IB coordinate mesh and interpolated back to a Cartesian mesh. Figure 5b shows the wavefield computed directly on a 3D Cartesian mesh, which is visibly identical to the wavefield in Figure 5a. Overall, these tests indicate that one may use the above 3D FDTD approach to compute impulse responses in generalized coordinate systems that are (nearly) as accurate as the corresponding Cartesian waveforms simulated by the equivalent Cartesian numerical FDTD scheme.

Finally, because I have specified the IB mesh analytically it is possible to examine the effects of variable grid spacing on the CFL

criterion and the ppw wavefield sampling. Assuming that the computational domain is uniform sampling along each coordinate axis (i.e., $\Delta\xi_1 = \Delta\xi_2 = \Delta\xi_3 = \Delta\xi_3$), the CFL criterion for the IB coordinate system may be derived according to the following equation:

$$\Delta t(\xi) \leq \frac{\Delta\xi}{v_{\xi}} \arg \min_{\xi_1, \xi_2, \xi_3} \left[2 + \frac{1}{(1 - 4\xi_3 + P(8\xi_3 - 4))^2 + 16\xi_3^2(\xi_3 - 1)^2 \left[\left(\frac{\partial P}{\partial \xi_1} \right)^2 + \left(\frac{\partial P}{\partial \xi_2} \right)^2 \right]} \right]^{-\frac{1}{2}}. \quad (31)$$

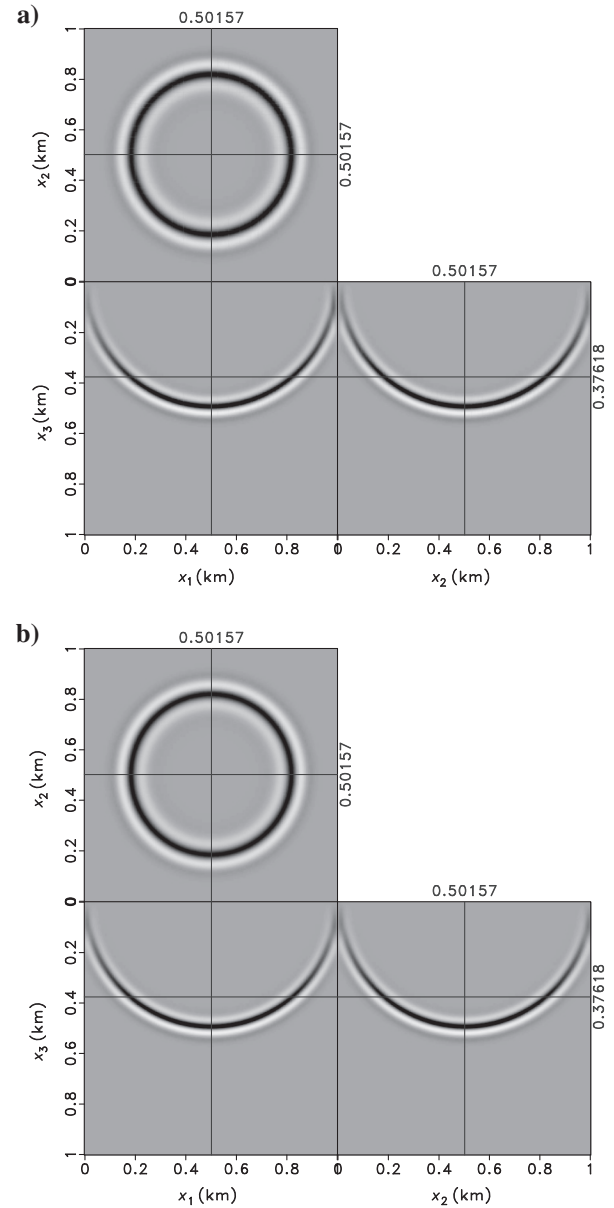


Figure 3. Impulse responses test results. (a) Impulse response in IB coordinates interpolated back to a Cartesian mesh. (b) Impulse response computed directly in a Cartesian coordinate system.

Similarly, the relative ppw sampling of the wavefield in the IB-coordinate mesh may be found by evaluating the expression in equation 26:

$$P(\xi) \approx \frac{v_\xi}{f} \frac{1}{\sqrt{2 + (1 - 4\xi_3 + P(8\xi_3 - 4))^2 + 16\xi_3^2(\xi_3 - 1)^2 \left[\left(\frac{\partial P}{\partial \xi_1} \right)^2 + \left(\frac{\partial P}{\partial \xi_2} \right)^2 \right]}} \quad (32)$$

Figure 6 shows the values of the $\Delta t(\xi)$ and $P(\xi)$ fields for the above IB coordinate system relative to those computed in a uniformly sampled Cartesian mesh of equivalent mean discretization interval. Figure 6a presents the relative spatial variability of the CFL criterion, which ranges between 0.47 and 1.15 times that of the Cartesian implementation. Mesh regions exhibiting grid compression result in finer spatial sampling and generally demand smaller time steps. Figure 6b similarly shows the relative change in ppw sampling, which range between 0.78 and 1.2 times that of the equivalent Cartesian implementation. Again, regions of grid compression lead to an increase in the ppw sampling of wavefields.

Semiorthogonal cylindrical coordinates

The second numerical test is a semiorthogonal cylindrical coordinate system, which is a useful mesh for performing 3D imaging and velocity inversion experiments using ultrasound data acquired on (nearly) cylindrical core plugs using piezoelectric and/or laser sources and receivers. Unlike the far more common orthogonal cylindrical coordinate system (Morse and Feshback, 1953), this coordinate system is semiorthogonal because one axis is orthogonal to the two other, mutually nonorthogonal axes. A semiorthogonal cylindrical coordinate system of radius $r_\xi \equiv \sqrt{\xi_1^2 + \xi_2^2} = 1$ can be specified by mapping equations

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \sqrt{1 - \frac{\xi_2^2}{2}} \\ \xi_2 \sqrt{1 - \frac{\xi_1^2}{2}} \\ \xi_3 \end{bmatrix}, \quad (33)$$

where the first two equations represent an analytic mapping of a circle to a square.

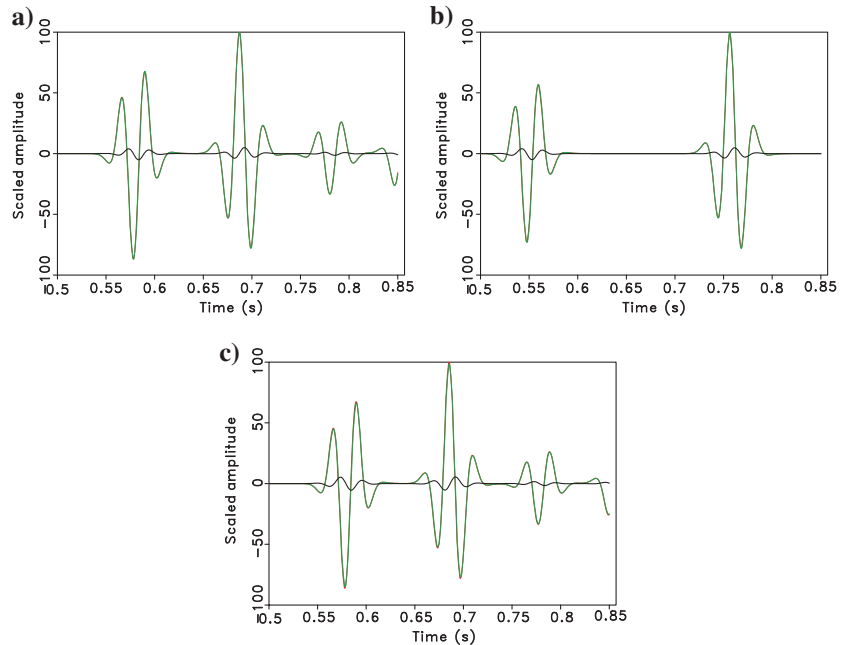
Figure 7a shows a 3D camera view of the coordinate system whereas Figure 7b-7d presents cross sections through the coordinate mesh taken at three constant planes. Figure 7b and 7c shows cuts through the $x_2 = 0$ and $x_1 = 0$ planes that form locally orthogonal rectilinear meshes. Figure 7d shows a cut through the $x_3 = 0$ plane that forms a disk with a (nearly) circular boundary. Note the presence of the four distorted mesh regions, which correspond to the four corners of the square in the $\xi_3 = 0$ plane. If the computational mesh were extended to $\xi_1, \xi_2 \in [-1, 1]$, these points would generate singularities in the metric tensor g_{ij} , creating a vanishing metric tensor determinant $|\mathbf{g}|$, and thereby causing numerical instability and requiring infinitely small time steps. Thus, for computational reasons I restrict the canonical domain to $\xi_1, \xi_2 \in [-0.995, 0.995]$. (The ξ_3 coordinate may be freely adjusted to account for the desired cylinder height.)

The analytic expression for the inverse metric tensor corresponding to the semiorthogonal cylindrical coordinate system in equation 33 can be derived after somewhat involved mathematical operations:

$$[g^{ij}] = \begin{bmatrix} \frac{(2 - \xi_1^2)(\xi_1^2 \xi_2^2 - \xi_1^2 - \xi_2^2 + 2)}{(\xi_1^2 + \xi_2^2 - 2)^2} & \frac{\xi_1 \xi_2 (\xi_1^2 - 2)(\xi_2^2 - 2)}{(\xi_1^2 + \xi_2^2 - 2)^2} & 0 \\ \frac{\xi_1 \xi_2 (\xi_1^2 - 2)(\xi_2^2 - 2)}{(\xi_1^2 + \xi_2^2 - 2)^2} & \frac{(2 - \xi_2^2)(\xi_1^2 \xi_2^2 - \xi_1^2 - \xi_2^2 + 2)}{(\xi_1^2 + \xi_2^2 - 2)^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (34)$$

Note that the semiorthogonality can be observed directly in the metric tensor of equation 34, where components $g^{13} = g^{23} = 0$, thus indicating zero “geometric coupling” between ξ_1 and ξ_2 with the

Figure 4. Selected traces extracted from propagation in a Cartesian coordinate system (red line), the IB coordinate system (green line) and the waveform difference (black line) for a source point located in the middle of the 3D mesh (i.e., at $\mathbf{x} = [0.5, 0.5, 0.0]$ km in Figure 2). (a) Wavefield traces extracted at $\mathbf{x} = [0.25, 0.25, 1.0]$ km. (b) Wavefield traces extracted at $\mathbf{x} = [0.5, 0.5, 1.0]$ km. (c) Wavefield traces extracted at $\mathbf{x} = [0.75, 0.75, 1.0]$ km.



ξ_3 direction. Likewise, the ζ^i coefficients for the first-order partial derivative operators may be derived analytically and are given by

$$[\zeta^i] = \begin{bmatrix} \frac{\xi_1(\xi_1^2\xi_2^2 - 3\xi_1^2\xi_2^4 - \xi_1^4 + 5\xi_1^2\xi_2^2 + 2\xi_1^2\xi_2^2 + 4\xi_1^2 - 8\xi_2^2 - 4)}{(\xi_1^2 + \xi_2^2 - 2)^3} \\ \frac{\xi_2(\xi_2^4\xi_1^2 - 3\xi_2^2\xi_1^4 - \xi_2^4 + 5\xi_2^2\xi_1^2 + 2\xi_2^2\xi_1^2 + 4\xi_2^2 - 8\xi_1^2 - 4)}{(\xi_1^2 + \xi_2^2 - 2)^3} \\ 0 \end{bmatrix}. \quad (35)$$

For the numerical tests described below, I use a computational mesh of dimension $N_\xi^3 = 400^3$ with a uniform sampling interval of $\Delta\xi =$

0.01 km and a constant-velocity of $v_\xi = 2 \text{ km s}^{-1}$. I inject a 20-Hz wavelet at the central grid point (i.e., $\xi = [0, 0, 0] \text{ km}$), which I define as the coordinate system origin. The corresponding CFL stability requirement for time step Δt can be found by evaluating the total derivative expressions in equation 22:

$$\Delta t = \frac{1}{v_\xi} \left[1 + \frac{2(\xi_1^2 - 2)}{(\xi_1^2 + \xi_1\xi_2 - 2)^2} + \frac{2(\xi_2^2 - 2)}{(\xi_2^2 + \xi_1\xi_2 - 2)^2} \right]^{-\frac{1}{2}}. \quad (36)$$

Figure 8a presents a plot of this function the corresponding maximum allowed time step Δt over the $[\xi_1, \xi_2] \in [-0.995, 0.995]$ coordinate range. There is significant spatial variability that ranges between $[1.0 \times 10^{-4} \leq \Delta t \leq 3.5 \times 10^{-3}] \text{ s}$. Note that the nonorthogonality of the coordinate system and the ensuing nonzero g^{12} and g^{13} terms leads to a scenario in which, in two corners, the coefficients work in favor of a larger Δt and two corners that demand a smaller Δt . Because specifying a stable FD scheme requires

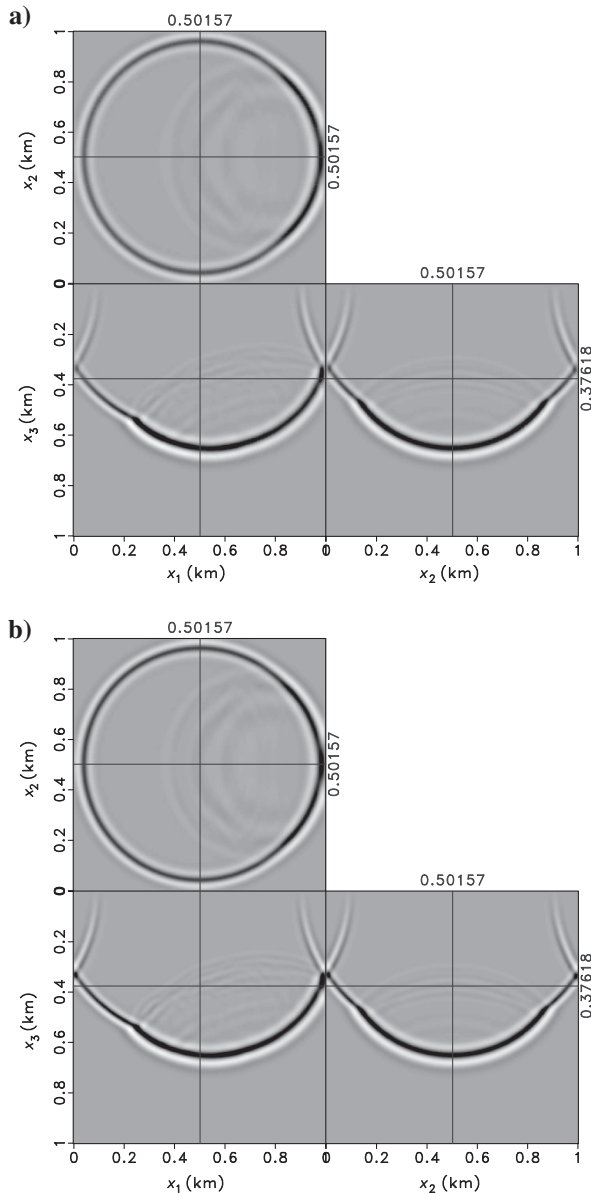


Figure 5. Wavefield propagation examples through the heterogeneous velocity model shown in Figure 2. (a) Wavefield computed through an IB mesh and interpolated back to Cartesian coordinates. (b) Wavefield computed directly in a Cartesian coordinate system.

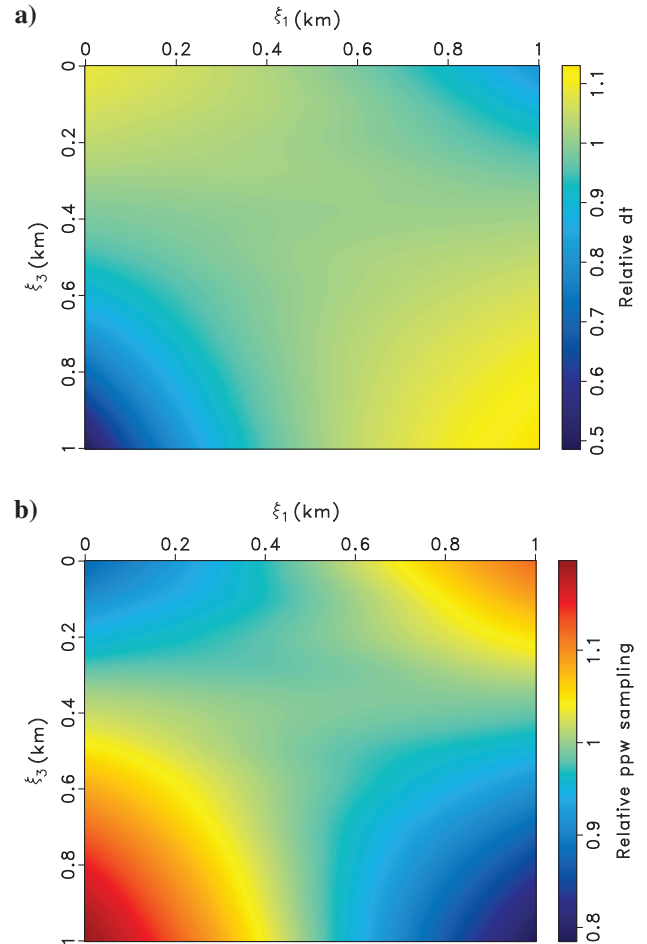


Figure 6. Effect of variable spatial sampling on the CFL criterion and ppw sampling of wavefields computed in 3D IB coordinates. (a) Map of relative spatial changes of the CFL criterion introduced by the IB geometry relative to a Cartesian coordinate system. (b) Relative ppw map in IB coordinates relative to that in Cartesian.

honouring the maximum time step allowed by the CFL stability condition, I chose the time step for this numerical example to be minimum value of $\Delta t = 1.0 \times 10^{-4}$ s. Values greater than this were tested but led to numerical instability as per the discussion above.

Figure 9 illustrates the utility of above 3D FDTD scheme by showing wavefield snapshots extracted at $x_3 = 0$ that were computed in the semiorthogonal cylindrical domain and subsequently interpolated back to Cartesian coordinates. The impulsive source was excited at the origin of a cylinder of unity radius. Based on the cylindrical symmetry in the $x_3 = 0$ plane, a point source excited at this location should experience the following series of events in the physical $\mathbf{x}$ -domain: (1) outward propagation as a circular wavefront, (2) reflection from the circular domain boundary with a $R = -1$ coefficient, (3) inward propagation as a collapsing circular wavefront, and (4) refocusing at the origin focus. Figure 9a shows wavefield propagating outward from the origin at time $t = 0.4$ s after source excitation. Note the presence of only a very modest amount of numerical dispersion noise located to the interior of the expanding wavefront. Figure 9b presents the wavefield at $t = 0.5$ s just as it reflects inward from the physical boundary (at radius $r_\xi = 1$). Figure 9c shows the inward propagating wavefield post boundary reflection at $t = 0.6$ s. Weak numerical artifacts associated with dispersive behavior are found now to the exterior of the collapsing wavefront. Finally, Figure 9d shows the wavefield at $t = 1.0$ s after it successfully collapses back to the focus located at the coordinate system origin.

3D topographic coordinates

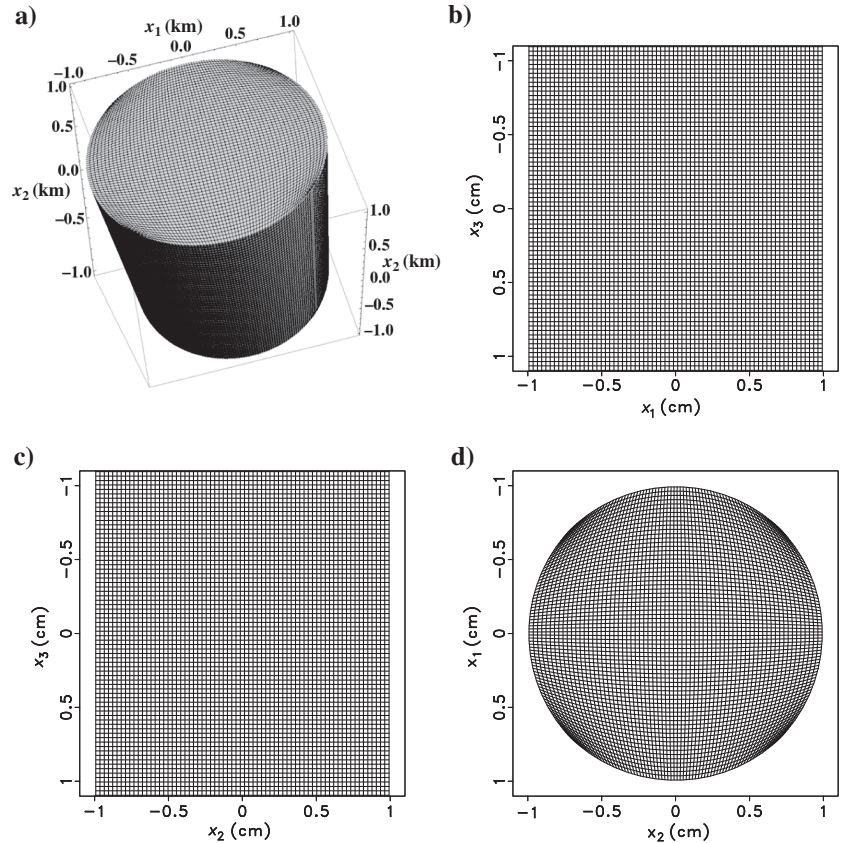
The third numerical test examines a canonical topography scenario in which surface relief is given by a Gaussian function of the type $\tau(\xi_1, \xi_2) = h e^{-a^2 \xi_1^2 - b^2 \xi_2^2}$. Figure 10 shows a representative topographic profile for vertical scaling parameters $h = 0.5$ km and Gaussian spread parameters $[a, b] = [2, 5/2]$ km⁻¹, which are also the parameters used in the numerical example discussed below. One may specify a semiorthogonal topographic coordinate system analytically through the use of linear Bézier functions in the ξ_3 coordinate:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ a\xi_3 + \tau(1 - \xi_3) \end{bmatrix}, \quad (37)$$

where a is vertical scaling parameter, $\xi_3 \in [0, 1]$, and ξ_1 and ξ_2 are allowed to freely vary. This mapping represents a linear blending between the surface topographic function $\tau(\xi_1, \xi_2, \xi_3 = 0)$ at the free surface and a uniform horizontal layer at a user-specified depth (i.e., $\xi_3 = a$).

The analytic transformation between the ξ - and $\mathbf{x}$ -domains again permits an analytic computation of the g^{ij} and ζ^i fields. The inverse metric tensor components g^{ij} are given by

Figure 7. Visualization of the 3D semiorthogonal cylindrical coordinate system showing $\mathbf{x}(\xi)$. (a) Extremal cylindrical surface with the coordinate system mesh overlain. (b) Cross section through the $x_2 = 0$ plane. (c) Cross section through the $x_1 = 0$ plane. (d) Cross section through the $x_3 = 0$ plane.



$$[g^{ij}] = \begin{bmatrix} 1 & 0 & \frac{\xi_3-1}{a-\tau} \frac{\partial \tau}{\partial \xi_1} \\ 0 & 1 & \frac{\xi_3-1}{a-\tau} \frac{\partial \tau}{\partial \xi_2} \\ \frac{\xi_3-1}{a-\tau} \frac{\partial \tau}{\partial \xi_1} & \frac{\xi_3-1}{a-\tau} \frac{\partial \tau}{\partial \xi_2} & \frac{1+(\xi_3-1)^2}{(a-\tau)^2} \left[\left(\frac{\partial \tau}{\partial \xi_1} \right)^2 + \left(\frac{\partial \tau}{\partial \xi_2} \right)^2 \right] \end{bmatrix}. \quad (38)$$

Inspection of the metric tensor demonstrates that the ξ_1 and ξ_2 directions are mutually orthogonal because the field component $g^{12} = 0$; however, these two axes are not orthogonal to ξ_3 as evidenced by nonzero g^{13} and g^{23} components. The only nonzero ζ^i coefficient is given by

$$\zeta^3 = \frac{\xi_3-1}{(a-\tau)^2} \left[2 \left(\frac{\partial \tau}{\partial \xi_1} \right)^2 + 2 \left(\frac{\partial \tau}{\partial \xi_2} \right)^2 + (a-\tau) \left(\frac{\partial^2 \tau}{\partial \xi_1^2} + \frac{\partial^2 \tau}{\partial \xi_2^2} \right) \right]. \quad (39)$$

The partial derivatives of the topographic profile τ defined above are

$$\begin{bmatrix} \frac{\partial \tau}{\partial \xi_1} \\ \frac{\partial \tau}{\partial \xi_2} \\ \frac{\partial^2 \tau}{\partial \xi_1^2} \\ \frac{\partial^2 \tau}{\partial \xi_2^2} \end{bmatrix} = \begin{bmatrix} -2a^2 \xi_1 \tau \\ -2b^2 \xi_2 \tau \\ (-2a^2 + 4a^4 \xi_1^2) \tau \\ (-2b^2 + 4b^4 \xi_2^2) \tau \end{bmatrix}. \quad (40)$$

Expressions for the CFL criterion and ppw sampling can be computed analytically as per above; however, they are omitted here for brevity.

I discretize the 3D topographic coordinate system in equation 37 using a $N_\xi^3 = 400^3$ mesh with a uniform sampling interval of $\Delta\xi = 7.5$ m along each coordinate axes. Figure 11 presents cross-section cuts of the numerical mesh as observed in Cartesian coordinates. Figure 11a shows the cross-section slice taken through the x_1 - x_3 -plane at the highest elevation point (i.e., at $x_2 = 0$ km). Figure 11b shows a similar cross-section cut through the x_2 - x_3 -plane at $x_1 = 0$ km. As illustrated in both panels, the total relief of the topographic perturbation is 0.5 km and is approximately 1.25- and 1.0-km wide in the $x_2 = 0$ km and $x_1 = 0$ km cross sections, respectively. Note also that the topographic coordinate system is piecewise smooth and has healed by bottom of the mesh located at $x_3 = 3.0$ km.

To illustrate the complexity generated by wavefield interaction with the free-surface topography, I simulate an acoustic impulse response from a point source located at $\xi = [0, 0, 0.275]$ km situated 0.225-km beneath the surface. For testing purposes, I again assume that the velocity model is a uniform $v_\xi = 2.0$ km s⁻¹. Figure 12 presents snapshots of the wavefield propagating through the semi-orthogonal topographic mesh after they have been interpolated back to the Cartesian $\mathbf{x}$ -domain. Figure 12a and 12b shows the wavefield (after interpolation to a Cartesian mesh) in the x_2 - x_3 -plane through the mesh center at $x_1 = 0$ km at $t = 0.24$ s and $t = 0.60$ s, respectively. In these snapshots, the most deeply propagated wavefield component is the downgoing wavefront that forms the expected hemispherical impulse response in the Cartesian $\mathbf{x}$ -domain. However, additional wavefield components have been generated by upward propagating energy that has reflected from the topographically influenced free surface (with a $R = -1$ reflection coefficient) and

generated the significant wavefield complexity (e.g., triplication, amplitude [de]focusing) evident in the $t = 0.6$ s panel. Note also that the wavefield reflects from topography regardless of the free-surface orientation and produces no staircase effects commonly observed in Cartesian FD implementations of wavefield propagation in topographic scenarios. The amplitudes also appear larger in the reflected wavefield components because the Gaussian-shaped topography focuses energy back toward the source location. Figure 12c and 12d shows the wavefield in the x_1 - x_3 -plane through the mesh origin at $x_2 = 0$ km again after interpolation to a Cartesian mesh at $t = 0.24$ s and $t = 0.60$ s, respectively. These panels indicate that the wavefield is more affected by shorter wavelength topography along this axis orientation, as is evidenced by stronger wavefield diffraction tails.

I present a second topographic example that simulates 3D acoustic wave propagation on a numerically specified semiorthogonal mesh constructed from a topographic relief map of the southern San

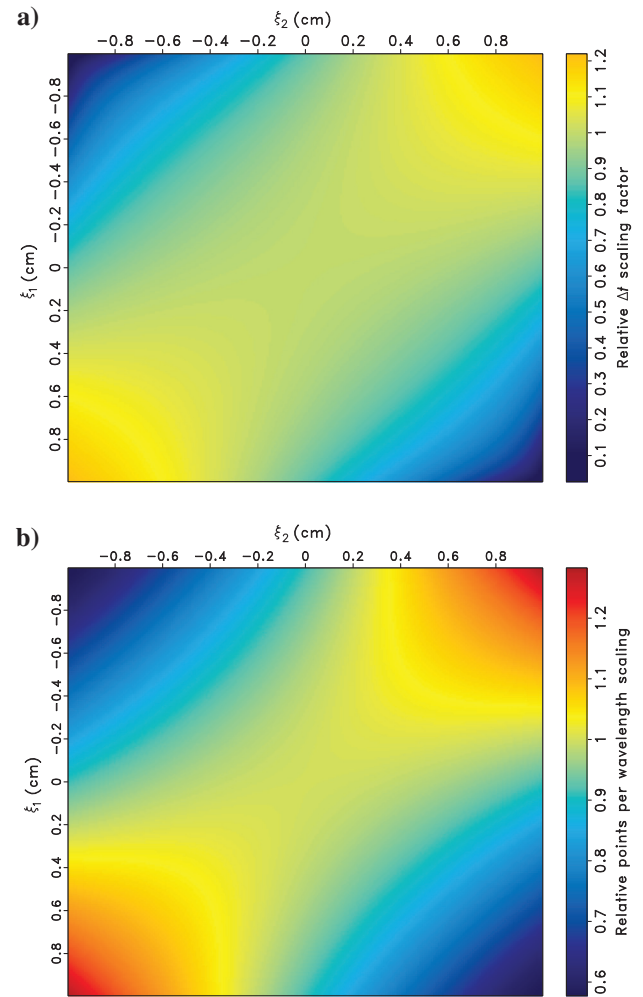


Figure 8. Relative spatial variability of numerical parameters introduced by semiorthogonal cylindrical geometry relative to a Cartesian coordinate system with the equivalent sampling. (a) Maximum allowed time step $\Delta t = \Delta t(\xi_1, \xi_2)$ according to modified CFL criterion in equation 20. (b) Ppw sampling in the physical domain $P(\xi)$ due to the generalized mesh.

Francisco Bay area. The elevation map presented in Figure 13 shows the fairly short wavelength and moderate amplitude variations, ranging from sea level at $x_3 = 0$ km to a height of $x_3 = 0.55$ km. I again construct a computational mesh conformal to the $\tau(\xi_1, \xi_2, \xi_3 = 0)$ topographic profile in Figure 13 using linear Bézier blending functions in the ξ_3 dimension according to coordinate mapping equations presented in equation 37. For numerical testing, I use a computational mesh of dimension $N_\xi^3 = 400^3$ and a uniform sampling interval of $\Delta\xi = 0.02$ km throughout the canonical ξ -domain. Figure 14a and 14b presents cross sections through the 3D computational mesh in the x_1 - x_3 and x_2 - x_3 -planes at $x_2 = 2.4$ km and $x_1 = 1.44$ km, respectively. The use of linear Bézier functions generates a nonorthogonal mesh throughout the computational domain that exhibits mesh roughness beneath topographic features (e.g., at $[x_1, x_2] = [1.25, 2.5]$ km).

Figure 15 presents 3D acoustic wavefield propagation results computed in the San Francisco Bay topographic mesh and interpolated back to Cartesian. For these simulations, I use a 10-Hz Ricker wavelet as an impulsive source located at coordinate $\xi = [4.0, 4.0, -0.25]$ km (i.e., 0.25-km beneath the free surface) and a uniform velocity model of $v_\xi = 2$ km s $^{-1}$. Figure 15 shows the wavefields in a cubeplot format in which the top face represents a depth slice in the x_1 - x_2 plane and the two lower faces represent cross sections in the x_1 - x_3 (left) and x_2 - x_3 (right) planes. The grey lines indicate locations and coordinates of the cross-section cuts. I also show the topographic surface for visualization purposes. Figure 15a

shows the wavefield at time $t = 0.5$ s. The impulse response at greater depths is hemispherical as expected; however, the waves reflecting from the free surface are influenced by the irregular topography, which leads to a moderate amount of scattering and

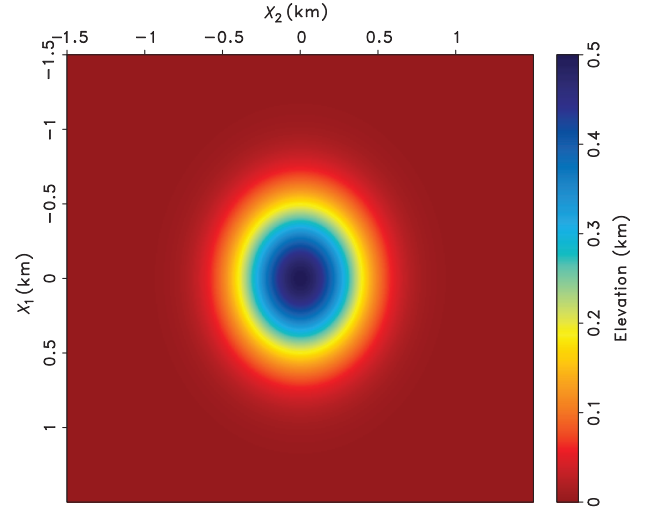
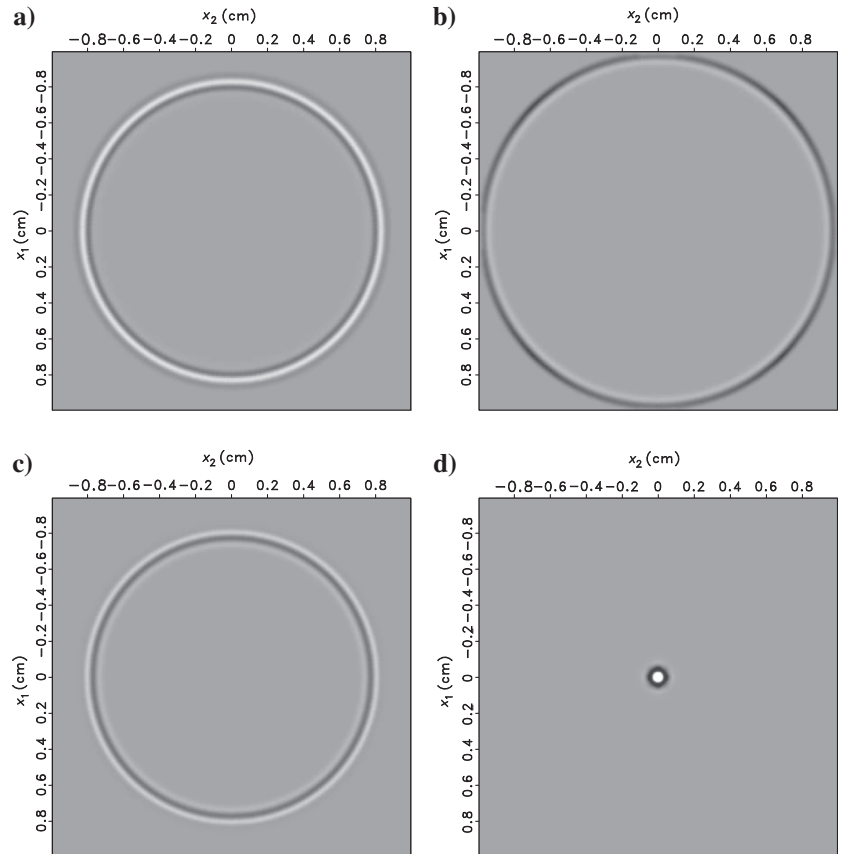


Figure 10. Surface topography defined by function $\tau(\xi_1, \xi_2, \xi_3 = 0) = \frac{1}{2}e^{-4\xi_1^2 - 25/4\xi_2^2}$.

Figure 9. Snapshots of the propagating wavefield interpolated back to a Cartesian coordinate systems in the $\xi_3 = 0$ plane. The source is located at $\mathbf{x} = \mathbf{0}$. The radius of the cylinder is unity. (a) Wavefield at $t = 0.4$ s. (b) Wavefield at $t = 0.5$ s during reflection at the cylindrical physical boundary. (c) Wavefield after boundary reflection at $t = 0.6$ s. (d) Wavefield at $t = 1.0$ s as it collapses back to the origin focus.



wavefield asymmetry. Figure 15b shows the wavefield evolved to $t = 1.0$ s. The nonhemispherical wavefield components are now significantly more complex and exhibit severe multipathing. Figure 15c and 15d shows the wavefields at $t = 1.5$ s and $t = 2.0$ s, respectively. These panels again show pronounced topographic scattering and wavefield multipathing, and thus illustrate the need for full-wave-

field propagators when handling these types of non-Cartesian geometry.

Overall, the above tests demonstrate the veracity of the analytic theory as well as the utility of the 3D FDTD wavefield propagation scheme detailed in the sections above to model 3D impulse responses and wavefield propagation in complex irregular non-

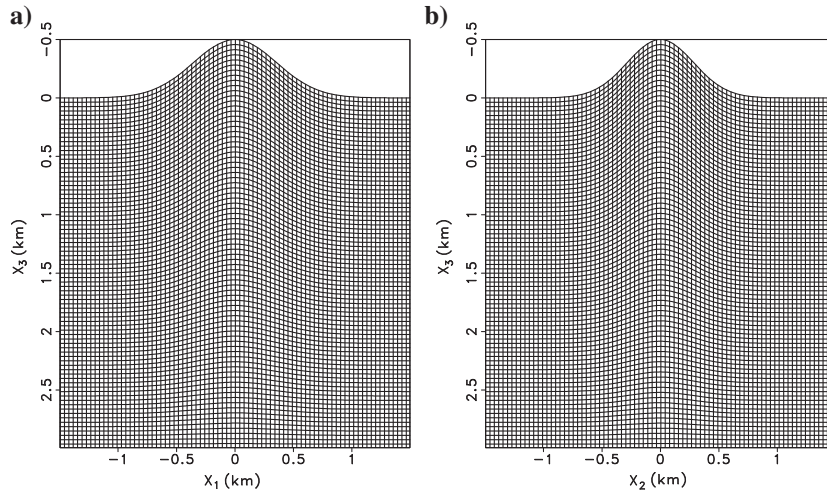


Figure 11. Cross sections of the semiorthogonal topographic mesh with 0.5 km elevation relief specified by equation 37. (a) Cross section in the x_1 - x_3 -plane taken at $x_2 = 0$ km. (b) Cross section in the x_2 - x_3 -plane taken at $x_1 = 0$ km.

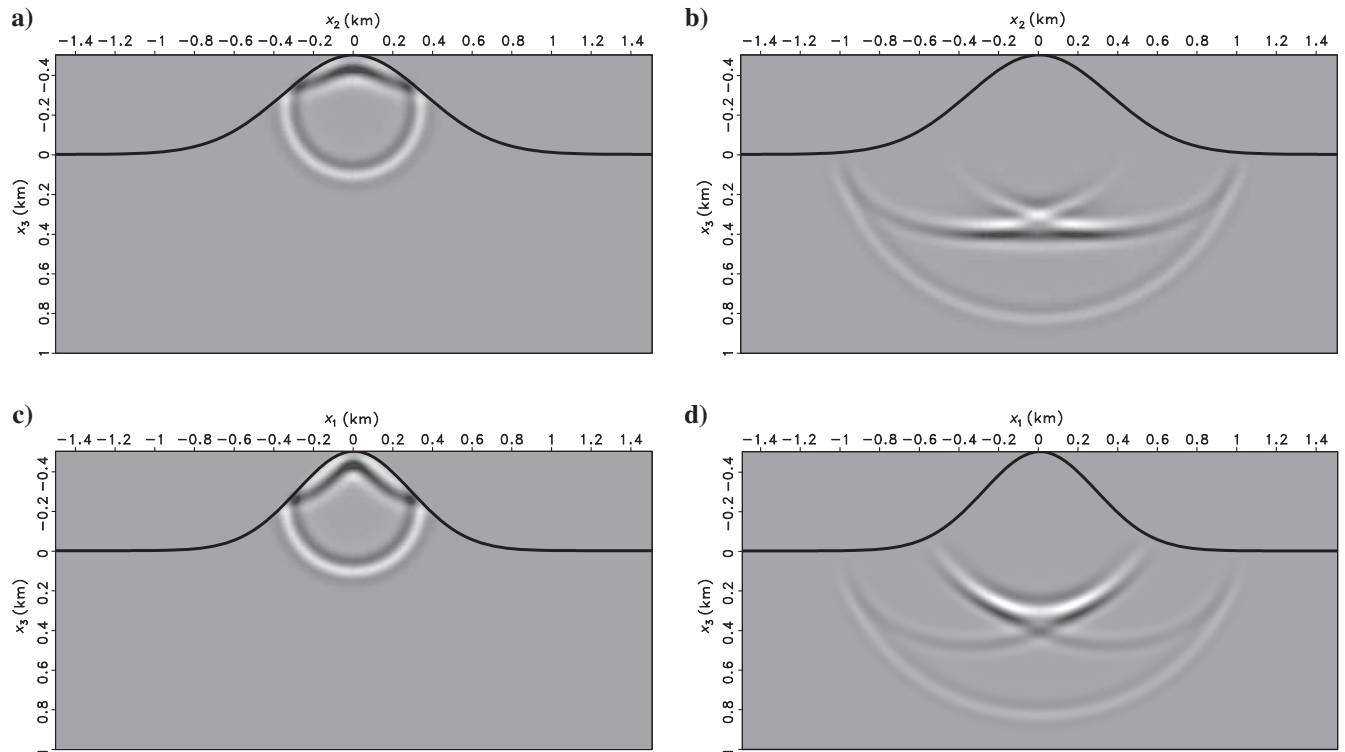


Figure 12. Wavefield propagation snapshots in the physical $\mathbf{x}$ -domain. (a and b) Wavefields in the x_1 - x_3 -plane at $x_2 = 0$ km interpolated to Cartesian coordinates (with the topographic surface indicated) at times $t = 0.24$ s and $t = 0.60$ s, respectively. (c and d) Wavefields in the x_2 - x_3 -plane at $x_1 = 0$ km interpolated to Cartesian coordinates (with the topographic surface indicated) at times $t = 0.24$ s and $t = 0.60$ s, respectively.

Cartesian geometries. Future work will address applying this scheme to scenarios involving non-Cartesian 3D RTM, FWI, and WEMVA, as well as developing the related 3D FD frequency-domain (FDFD) methodology useful for frequency-domain FWI applications.

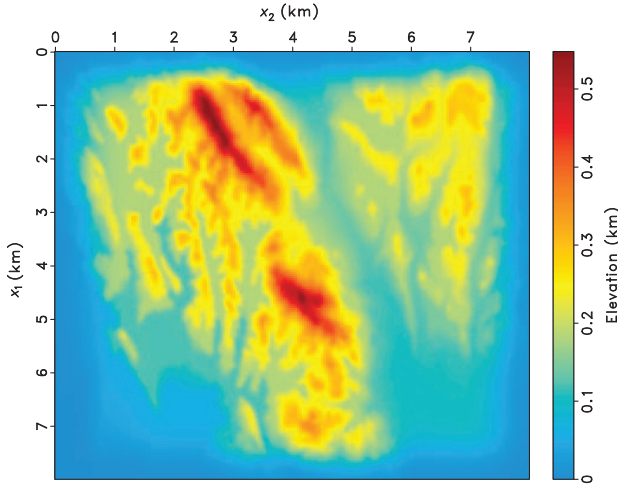


Figure 13. Topographic relief in the southern San Francisco Bay area forming surface $\tau(\xi_1, \xi_2, \xi_3 = 0)$.

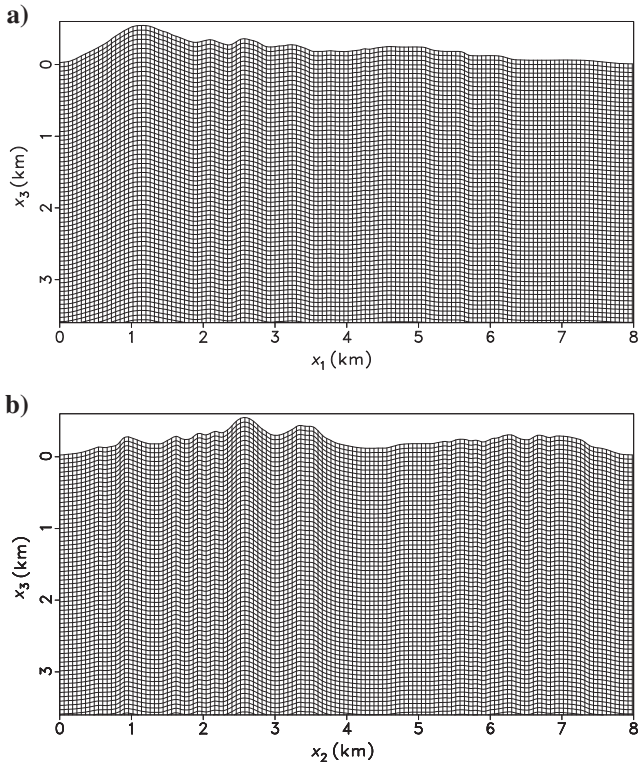


Figure 14. 2D cross sections through the topographic coordinate system constructed using linear Bézier functions; (a) x_1 - x_3 -plane at coordinate $x_2 = 2.4$ km; (b) x_2 - x_3 -plane at coordinate $x_1 = 1.44$ km.

DISCUSSION

As mentioned above, using the generalized 3D FDTD approach introduces additional memory overhead due to the requirements of handling of spatially varying g^{ij} and ζ^i fields. Wavefield modeling on a N_ξ^3 model domain requires holding a total of approximately $14N_\xi^3$ floating point numbers in memory, in which $9N_\xi^3$ are required for the g^{ij} and ζ^i fields and for the $5N_\xi^3$ for the wavefields, velocity model and image volume. When using this approach for 3D RTM or FWI applications the memory requirements increase to $16N_\xi^3$ due to the introduction of two additional receiver wavefields. Thus, the memory complexity of the generalized 3D FDTD approach when used for RTM or FWI applications will be approximately $2.3 \times$ that of a 3D Cartesian FDTD approach.

A second potential limitation arises due to the increased complexity of FD stencil applied in Figure 1a. As discussed above, the 73-point FD stencil proposed above is $2.92 \times$ larger than the 25-point star commonly used for $O(\Delta\xi^8)$ 3D FDTD implementations shown in Figure 1b. This implies that there should be a moderate computational overhead when following the above approach. However, the corresponding increase in computational complexity may be reduced by recognizing that neighboring stencils share a large number of points and thus neighboring wavefield values can be reused during FD stencil evaluation. To address this issue, I implemented the above algorithm in CUDA so as to exploit the massive parallelism and memory hierarchy afforded by the GPU architecture. I adapted the GPU-based stencil approach described in Micikevicius (2009) and Weiss and Shragge (2013), in which wavefield values are prefetched and stored in $24 \times 24 \times 9$ sized blocks in the L1 cache for use in each thread block independently computing the FD stencils (thereby avoiding a significant number of redundant and relatively very slow calls to/from the GPU's global memory). The GPU architecture allows users to exploit the relatively very fast and low-latency access between the compute cores and the L1 cache and to code highly coalesced memory access patterns during the required stencil evaluations.

Figure 1c shows the 3200 grid points required to collectively evaluate the generalized 3D FDTD stencil for a $16 \times 16 \times 1$ section of the mesh. This necessarily includes a four-point halo region on each grid face (shown in green) leading to a total $24 \times 24 \times 9$ block (less a few unneeded points). Figure 1d shows the 2688 grid points required to evaluate the same section of the 3D mesh but for the Cartesian 25-point FD star discussed above. As illustrated pictorially, reading wavefields into the L1 cache en masse greatly reduces read redundancy and requires prefetching only $1.2 \times$ additional wavefield values from the GPU's global memory.

The Cartesian and the generalized coordinate 3D FDTD propagation codes were both benchmarked on a model of dimensions $N_\xi^3 = 400^3$ for 1000 time steps using a NVIDIA K10 GPU with 4 GB GPU memory and 1536 (CUDA) cores (i.e., on one of the two K10 GPU devices). The Cartesian code ran in 30 s, where as the generalized coordinate code completed in 87 s, or approximately $2.92 \times$ slower (interestingly, in proportion to the increase in required number of stencil points). Although the Cartesian approach is fairly optimized in terms of the GPU implementation strategy (Micikevicius, 2009), additional optimizations are yet to be implemented on the generalized 3D FDTD code that will help reduce the apparent increase in computational complexity from 2.9 to $1.2 \times$. A detailed discussion on GPU implementation, though, remains outside of the scope of this paper.

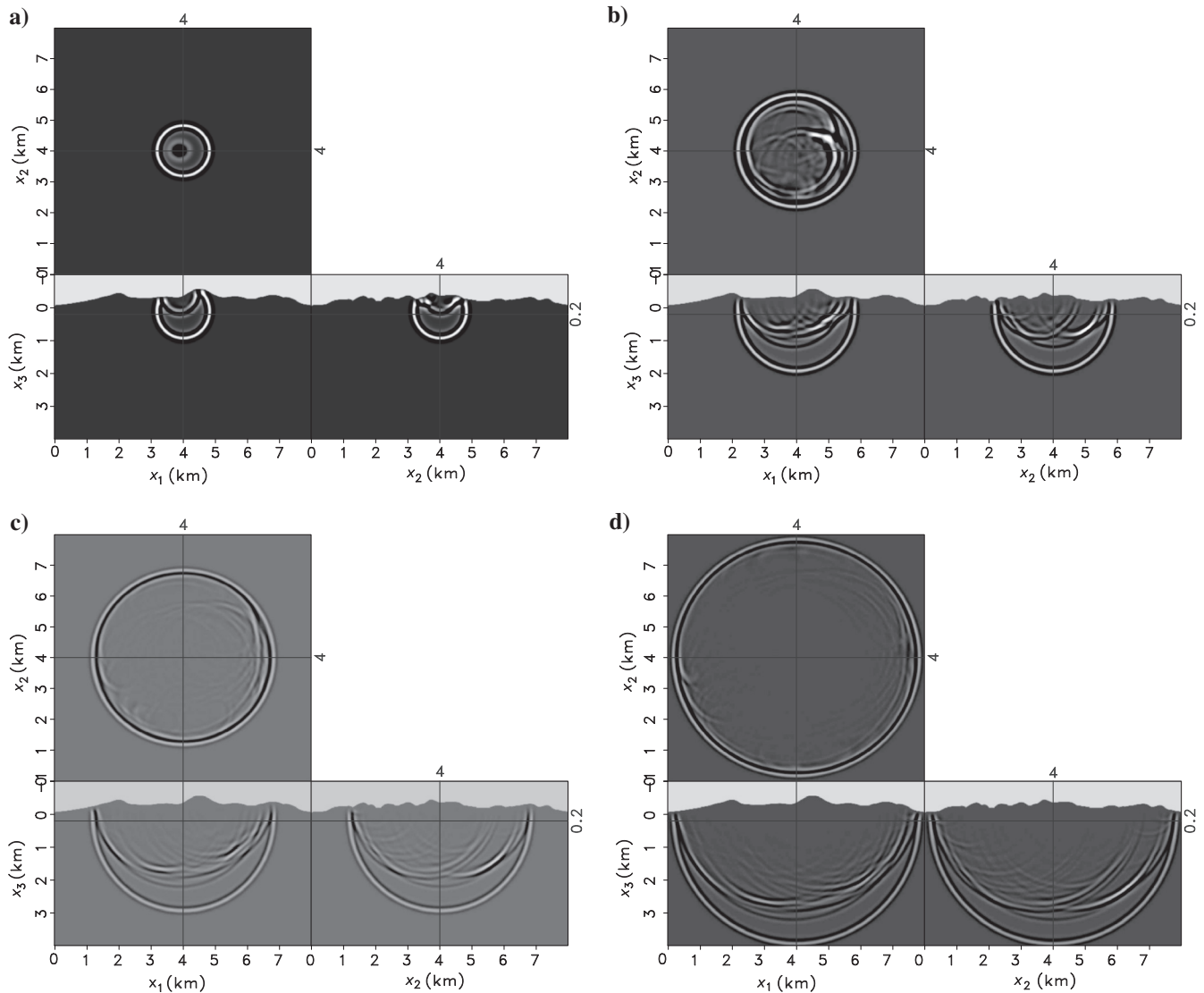


Figure 15. Time snapshots of the propagating 3D wavefield calculated in the ξ -domain and interpolated back to the $\mathbf{x}$ -coordinate domain. The background has been colored to illustrate the location of the free surface. (a) $t = 0.5$ s. (b) $t = 1.0$ s. (c) $t = 1.5$ s. (d) $t = 2.0$ s.

CONCLUSIONS

Applying a 3D FDTD wave propagation strategy on generalized structured meshes is feasible using a combination of coordinate mappings and differential geometry. The two key changes relative to a Cartesian 3D FDTD methodology are the introduction of first-order and mixed second-order derivatives as well as geometric scaling factors that account for the spatially varying geometry and its corresponding effects when simulating solutions to the 3D acoustic wave equation. The examples demonstrate that the analytic 3D acoustic wave equations can be derived for various canonical imaging problems, and that stable and accurate solutions of these equations can be found using a generalized coordinate 3D FDTD strategy. I demonstrate that analytic sufficiency expressions can be developed for the stability criterion and ppw sampling. Finally, the generalized 3D FDTD approach provides accurate 3D impulse responses and thus can be used as the computational kernel for non-Cartesian 3D imaging and velocity inversion experiments such as

RTM, FWI, and WEMVA, and potentially, related 3D FDFD FWI approaches.

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